

TECHNIQUE MASTERING EXERCISES

L - ANTIDERIVATIVES

1. $f'(x) = x^8$

Antw / Ans: $\int f'(x)dx = \int x^8 dx = \frac{x^9}{9} + k = \frac{1}{9}x^9 + k$

2. $f'(x) = \frac{1}{x^6}$

Antw / Ans: $\int f'(x)dx = \int x^{-6} dx = \frac{x^{-5}}{-5} + k = -\frac{1}{5x^5} + k$

3. $f'(x) = x^{5/7}$

Antw / Ans: $\int f'(x)dx = \int x^{5/7} dx = \frac{x^{12/7}}{\frac{12}{7}} + k = \frac{7}{12}\sqrt[7]{x^{12}} + k$

4. $f'(t) = \sqrt[3]{t^2}$

Antw / Ans: $\int f'(t)dt = \int t^{2/3} dt = \frac{t^{5/3}}{\frac{5}{3}} + k = \frac{3}{5}\sqrt[3]{t^5} + k$

5. $f'(x) = \frac{4}{\sqrt{x}}$

Antw / Ans: $\int f'(x)dx = \int 4x^{-1/2} dx = 4\frac{x^{1/2}}{\frac{1}{2}} + k = 8\sqrt{x} + k$

6. $f'(x) = \frac{1}{2x^3}$

Antw / Ans: $\int f'(x)dx = \int \frac{1}{2x^3} dx = \frac{1}{2} \int x^{-3} dx = \frac{1}{2} \frac{x^{-2}}{-2} + k = -\frac{1}{4x^2} + k$

7. $f'(x) = x^3 \sqrt{x}$

Antw / Ans: $\int f'(x)dx = \int x^{7/2} dx = \frac{x^{9/2}}{\frac{9}{2}} + k = \frac{2}{9}\sqrt{x^9} + k$

8. $f'(x) = x^3 - 2x + 7$

Antw / Ans: $\int f'(x)dx = \int (x^3 - 2x + 7)dx = \frac{x^4}{4} - 2\frac{x^2}{2} + 7x + k = \frac{1}{4}x^4 - x^2 + 7x + k$

9. $f'(x) = x^{-3} + \sqrt{x} - 3x^{1/4} + x^2$

Antw / Ans: $\int f'(x)dx = \int (x^{-3} + x^{1/2} - 3x^{1/4} + x^2)dx$
 $= \frac{x^{-2}}{-2} + \frac{x^{3/2}}{\frac{3}{2}} - 3\frac{x^{5/4}}{\frac{5}{4}} + \frac{x^3}{3} + k = -\frac{1}{2x^2} + \frac{2}{3}\sqrt{x^3} - \frac{12}{5}\sqrt[4]{x^5} + \frac{1}{3}x^3 + k$

10. $f'(u) = u^{2/3} - 4u^{1/5} + 4$

Antw / Ans: $\int f'(u)du = \int (u^{2/3} - 4u^{1/5} + 4)du =$
 $\frac{u^{5/3}}{\frac{5}{3}} - 4\frac{u^{6/5}}{\frac{6}{5}} + 4u + k = \frac{3}{5}\sqrt[3]{u^5} - \frac{10}{3}\sqrt[5]{u^6} + 4u + k$

11. $f'(x) = \frac{7}{x^{3/4}} - \sqrt[3]{x} + 4\sqrt{x}$

Antw / Ans: $\int f'(x)dx = \int \left(7x^{-3/4} - x^{1/3} + 4x^{1/2}\right)dx = 7\frac{x^{1/4}}{\frac{1}{4}} - \frac{x^{4/3}}{\frac{4}{3}} + 4\frac{x^{3/2}}{\frac{3}{2}} + k$
 $= 28\sqrt[4]{x} - \frac{3}{4}\sqrt[3]{x^4} + \frac{8}{3}\sqrt{x^3} + k$

12. $f'(x) = x(1 + x^3)$

Antw / Ans: $\int f'(x)dx = \int (x + x^4)dx = \frac{x^2}{2} + \frac{x^5}{5} + k = \frac{1}{5}x^5 + \frac{1}{2}x^2 + k$

13. $f'(t) = (1 + t^2)(2 - t)$

Antw / Ans: $\int f'(t)dt = \int (-t^3 + 2t^2 - t + 2)dt$
 $= -\frac{t^4}{4} + 2\frac{t^3}{3} - \frac{t^2}{2} + 2t + k = -\frac{1}{4}t^4 + \frac{2}{3}t^3 - \frac{1}{2}t^2 + 2t + k$

14. $f'(x) = x^{1/3}(2 - x)^2$

Antw / Ans: $\int f'(x)dx = \int (4x^{\frac{1}{3}} - 4x^{\frac{4}{3}} + x^{\frac{7}{3}})dx = 4\frac{x^{\frac{4}{3}}}{\frac{4}{3}} - 4\frac{x^{\frac{7}{3}}}{\frac{7}{3}} + \frac{x^{\frac{10}{3}}}{\frac{10}{3}} + k$
 $= 3\sqrt[3]{x^4} - \frac{12}{7}\sqrt[3]{x^7} + \frac{3}{10}\sqrt[3]{x^{10}} + k$

15. $f'(x) = \frac{1}{x^2} - \cos x$

Antw / Ans: $\int f'(x)dx = \int (x^{-2} - \cos x)dx = \frac{x^{-1}}{-1} - \sin x + k = -\frac{1}{x} - \sin x + k$

16. $f'(\theta) = 4\sin \theta + 2\cos \theta$

Antw / Ans: $\int f'(\theta)d\theta = \int (-4(-\sin \theta) + 2\cos \theta)d\theta = -4\cos \theta + 2\sin \theta + k$

17. $f'(y) = e^y$

Antw / Ans: $\int f'(y)dy = \int e^y dy = e^y + k$

18. $f'(x)dx = xe^{x^2}$

Antw / Ans: $\int f'(x)dx = \frac{1}{2} \int e^{x^2}(2x)dx = \frac{1}{2}e^{x^2} + k$

19. $f'(x) = \frac{2}{x}$

Antw / Ans: $\int f'(x)dx = 2 \int \frac{1}{x}dx = 2\ln x + k$